ZCPSC
Zimbabwe Commodity Price Stress Classifier

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Research for a more food secure Zimbabwe

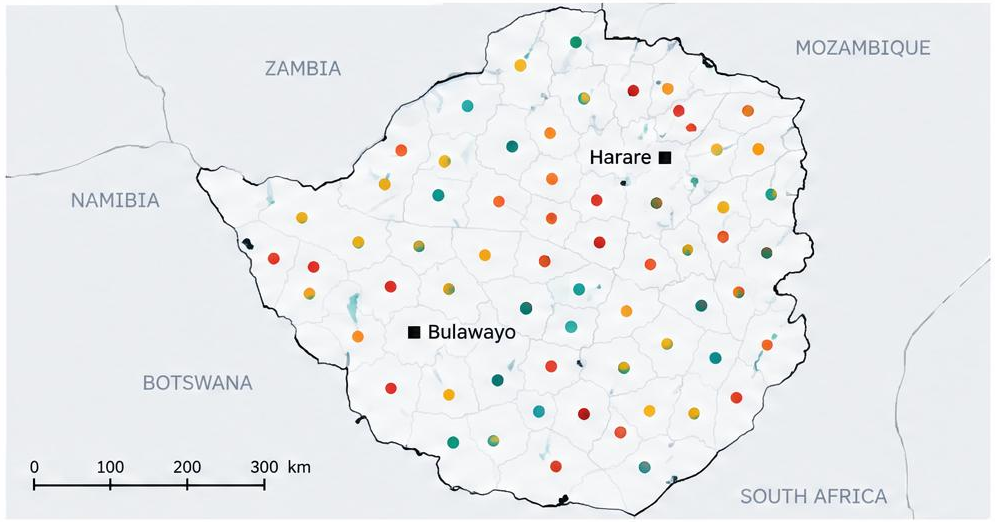
ZCPSC Overview

Next-month relative commodity price stress prediction for Zimbabwean districts

Districts monitored	Commodities	Prediction horizon	Model status
81	18	1 month	Ready

Predicted stress overview

Each point represents a district. Colour indicates the predicted probability of commodity price stress in the next month.



Recent prediction summary

Number of districts in each predicted stress category for next month.

High probability	18
Moderate probability	27
Lower probability	36

Out of 81 monitored districts.

How the system works

A simple three-step workflow from input to insight.

1. **Enter information at month T**
   Provide district, commodity and relevant information for the current month.
2. **Model predicts month T+1**
   Our machine learning model estimates the probability of price stress next month.
3. **Investigate results**
   Explore district-level predictions, compare commodities and understand drivers using model explanations.

⚠ Academic decision-support prototype. Not an official food-security assessment, IPC classification or guaranteed forecast.

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Next-Month Prediction

Use information available at month T to estimate relative commodity-price stress in month T+1

Prediction month: **October 2024**

Target month: **November 2024**

Model: **XGBoost Option B**

Single observation input

District	Chitungwiza	Recent SPI lag 1	1.08
Commodity	Maize grain	Rainfall 3-month total	184 mm
Currency	USD	Population	356,840
Prediction month	October 2024		

Predict Next Month Stress

Prediction result

Elevated relative price stress

72.4%

predicted probability

For target month: **November 2024**

Low — Moderate — High

Based on information available at month T (October 2024), the model estimates a 72.4% probability of elevated relative commodity-price stress in month T+1 (November 2024).

Top model contributions

Feature	Relative contribution	Value
SPI lag 1	42%	1.08
Rainfall 3-month total	28%	184 mm
Commodity identity	18%	Maize grain

Batch prediction

Upload CSV template | Download template | Validate CSV

12 rows uploaded | 10 valid rows | 2 rows need attention

Disclaimer: Predictions are model-based estimates of relative price stress and should be used for analytical and planning purposes only. They are not guarantees and may be affected by data limitations and unforeseen events.

CSV Batch Prediction

Upload multiple district–commodity observations and predict relative price stress for month T+1

Drag and drop CSV file here

Upload a completed Option B prediction template

Download CSV template | Browse files

Validation summary

Rows uploaded	Valid rows	Rows needing attention
24	21	3

Validation details

Row	District	Commodity	Status	Message
1	Chitungwiza	Maize grain	Valid	—
2	Gweru	Sorghum	Valid	—
3	Mutare	Wheat	Needs attention	Missing rainfall feature

Predict Valid CSV Rows

Prediction output preview

Observation ID	District	Target month	Predicted class	Probability
OBS-0001	Chitungwiza	2024-11	Low	0.78
OBS-0002	Gweru	2024-11	Moderate	0.62
OBS-0003	Mutare	2024-11	High	0.81

Download predictions CSV | Download validation errors

Predictions are model-based estimates and should be used for analysis and planning support, not as the sole basis for decision-making.

ZCPSC
Zimbabwe Council for Price Stability and Cooperation

ZCPSC User

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Explore Data

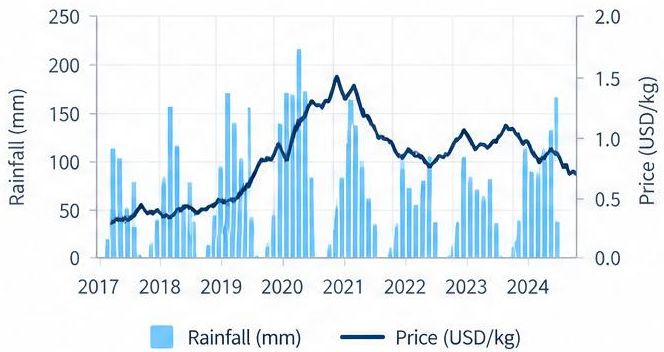
Inspect market, rainfall and contextual information used by the predictive system

District	Commodity	Currency	Date range
All	All	USD	2017 – 2024

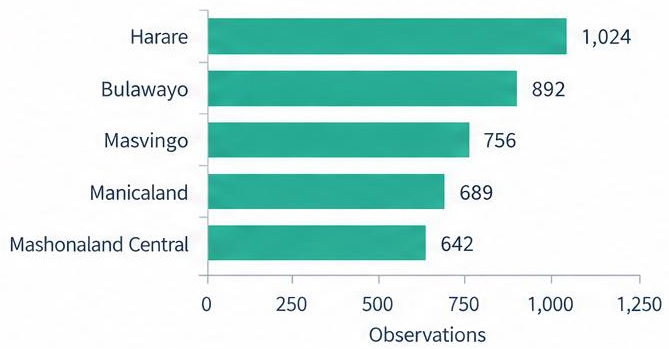
Spatial Price Index by month



Rainfall and recent price context



Observations by district



Data coverage

District	Commodity	First month	Last month	Valid observations
Harare	Maize	2017-01	2024-12	1,024
Bulawayo	Maize	2017-01	2024-12	892
Masvingo	Groundnuts	2017-01	2024-12	756
Manicaland	Beans	2017-01	2024-12	689
Mashonaland Central	Sorghum	2017-01	2024-12	642

Exploration describes observed historical data; it is not the same as a next-month prediction.

Source: ZCPSC market, rainfall and auxiliary datasets (2017 – 2024).
Note: Data coverage and values may vary by district and commodity due to data availability and quality checks.

Stable Markets
Stronger Communities

Zimbabwe Council for Price Stability and Cooperation (ZCPSC)
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Model Explanations

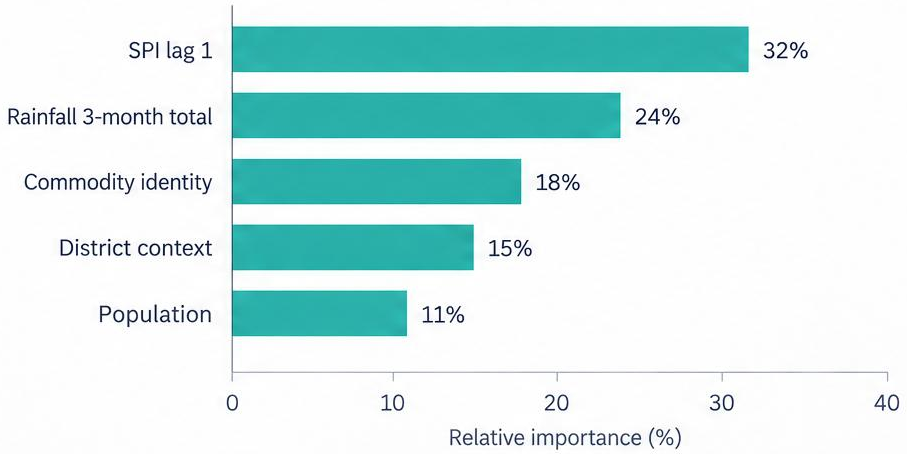
Tuesday, 15 October 2024

Understand which features contributed to the fitted model predictions

Model: XGBoost Option B | District: Chitungwiza | Commodity: Maize grain | Target month: November 2024

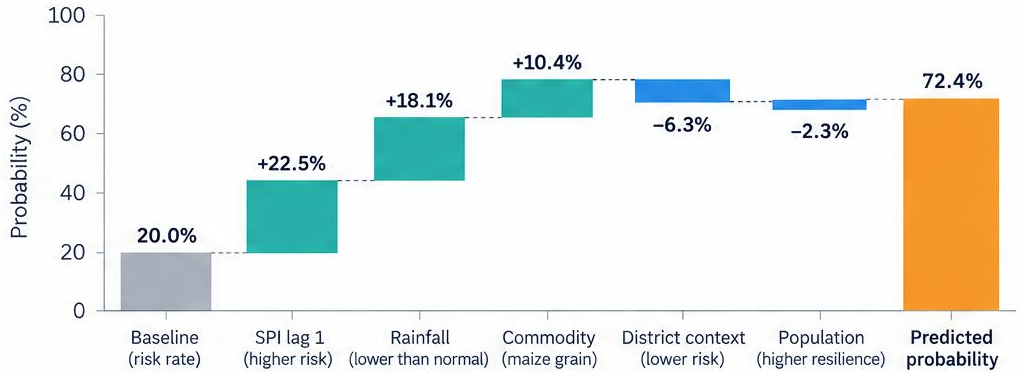
Global feature importance

Relative importance of features in the fitted model (higher values indicate greater influence on predictions).



Local explanation

Feature contributions for this specific prediction (relative to a baseline).



Predicted probability: 72.4%

Feature contribution explains the fitted model prediction; it does not prove causation.

Interpretation guidance

Use explanations to investigate
Identify which factors are driving predictions and explore the underlying data.

Do not infer causation
Feature contributions show association with the model prediction, not cause and effect.

Compare with official evidence
Validate insights using government reports, field assessments and other trusted sources.

Disclaimer: Model explanations are based on the fitted model and available data. They are for informational use only and should be interpreted alongside official statistics, expert knowledge and on-the-ground evidence. They do not establish causation and should not be used as the sole basis for decisions.

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Methodology

How the next-month prediction system is constructed and evaluated

1 **Data integration**
Market prices, rainfall, population and geography

2 **Target construction**
SPI > 1.25 and target shifted to month T+1

3 **Model validation**
Forward-chaining temporal validation and leakage controls

4 **Model explanation**
SHAP feature contributions and transparent communication

Method workflow

Information available at T → Feature pipeline → Model → Prediction for T+1

Key definition

$$X_t \rightarrow Y_{(t+1)}$$

Predict relative commodity-price stress in the following month.

Models compared

- DummyClassifier baseline
- Logistic Regression
- XGBoost

This system is for academic research purposes only. Predictions are not a substitute for professional advice or decision-making.

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Price Trends
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Key Drivers (SHAP)
Commodity Analysis
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Limitations and Responsible Use

Important information for interpreting model-based predictions

Use predictions as supporting evidence

This prototype does not ireplace official food-security assessment, expert judgement or field verification.

Data limitations

- Uneven reporting coverage across commodities, markets and time periods.
- Based on historical public data, which may be incomplete or subject to revision.
- Possible missing values and data quality issues.

Model limitations

- Class imbalance (relatively few high price-stress events).
- Potential distribution shift if future conditions differ from historical patterns.
- Predictions include uncertainty and should be interpreted probabilistically.
- No guaranteed forecast — model performance may vary over time and across commodities.

Interpretation limits

- Price stress is not the same as household food insecurity.
- SHAP explains model behaviour, not causation.
- No IPC phase assignment — this dashboard does not classify areas into IPC phases.

Responsible-use statement

This dashboard provides model-based predictions of next-month relative commodity-price stress. It is an academic decision-support prototype. It does not constitute an official food-security assessment, an IPC classification or a guaranteed forecast.

Before using results

- [ ] Review data coverage
- [ ] Check prediction month and target month
- [ ] Consult official evidence